St. Philomena's College (Autonomous), Mysore

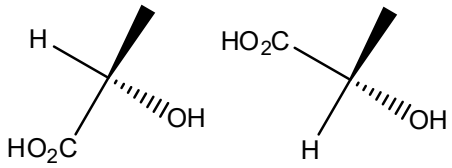
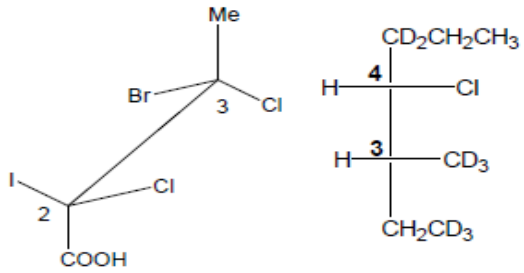
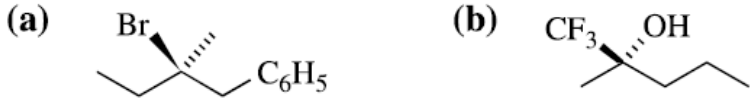
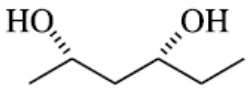
PG Department of Studies and Research in Chemistry

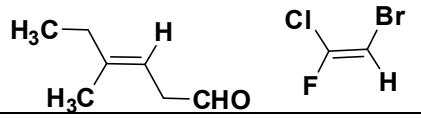
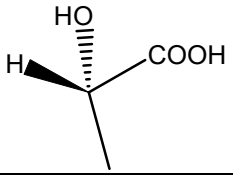
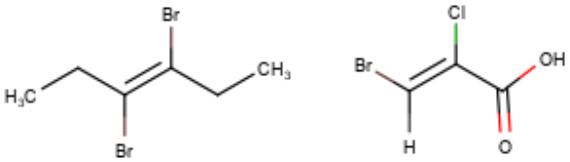
Question Bank (Revised Curriculum 2020 onwards)

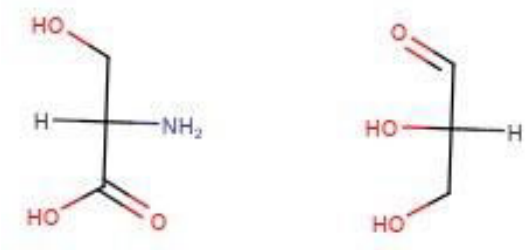
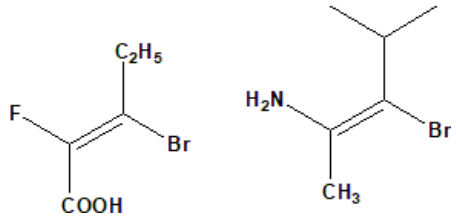
First Year- First Semester (2020 -21 Batch)

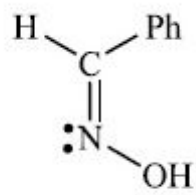
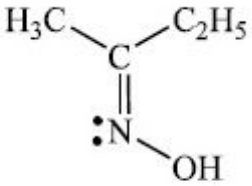
Course Title (Paper Title):

STEREOCHEMISTRY, REACTION MECHANISMS AND HETEROCYCLIC CHEMISTRY **QP Code: 85123**

Unit/ Module	SI. No	Questions	Marks
1	1.	Assign absolute configuration and identify the relation between given molecules 	2
1	2.	Assign the absolute configuration of the following molecules 	2
1	3.	Give the Newman projection of (2R, 3R)-2-hydroxy-3-chlorobutanoic acid.	2
1	4.	Identify the chiral center and provide the absolute configuration (R/S) of the given molecules. 	2
1	5.	Identify the chiral centers and provide the absolute configuration (R/S) of the given molecule. 	2

1	6.	What are Stereo specific reactions? Give an example.	2
1	7.	Assign E and Z configuration to the following:  	2
1	8.	Write the stereochemical structure of D and L glyceraldehydes	2
1	9.	Differentiate mesocompounds and Racemic mixtures.	2
1	10.	Draw a Newman projection of the most stable conformation of 2-methylpropane.	2
1	11.	Draw the most stable conformation of cis-1-ethyl-4-isopropylcyclohexane.	2
1	12.	Draw the most stable conformation of cis-1-ethyl-3-methylcyclohexane.	2
1	13.	Define asymmetric synthesis?	2
1	14.	Illustrate stereoselective reaction?	2
1	15.	Illustrate is enantioselective synthesis	2
1	16.	Define stereospecific and regioselective reaction?	2
1	17.	Assign R/S configuration for 1. D- Glyceraldehyde 2. L-Serine	2
1	18.	Define enantiomers with an example.	2
1	19.	Define Diastereomers with an example.	2
1	20.	Convert the given structure to Fischer projection and assign the configuration.  	2
1	21.	Write E/Z configuration of the following Compounds  	2
1	22.	Differentiate between Enantiomers and Diastereomers.	2
1	23.	What is meant by molecular dissymmetry?	2

1	24.	Write R/S Configuration for following compounds 	2
1	25.	Draw the possible conformations of n-butane. Identify the conformer.	2
1	26.	Mention the condition for optical Isomerism.	2
1	27.	What are racemic mixtures? Why racemic mixture does not exhibit optical activity?	2
1	28.	In case of 1,4-dihydroxycyclohexaneboat conformation is more stable than chair conformation . Why?	2
1	29.	Draw the Sawhorse and Newman Projection formulae of 2-bromo-3-hydroxy-butane.	2
1	30.	Illustrate diastereoselective reaction with suitable example.	2
1	31.	Sketch various conformations of 1, 3-dimethyl cyclohexane.	2
1	32.	Assign the configuration of the following E or Z using CIP Method. 	2
1	33.	Distinguish Erythrose from threose.	2
1	34.	Write the most stable conformation of n-propyl chloride and ethylene glycol.	2
1	35.	Write E/Z notation for 3-Chloro-4-methyl-3- Hexene.	2
1	36.	Draw staggered Newman projection formula of n-butane.	2
1	37.	What are stereoselective reaction? Give an example.	2
1	38.	What are meso compounds? Give an example.	2
1	39.	What is regiospecific reaction? Give an example.	2
1	40.	Draw the most stable conformation of cis-1,3-dimethylcyclohexane.	2
1	41.	Draw the gauche Newman Projections for $CH_3CHBrC_2H_5$.	2
1	42.	Account for the optical activity observed in 2-butanol.	2

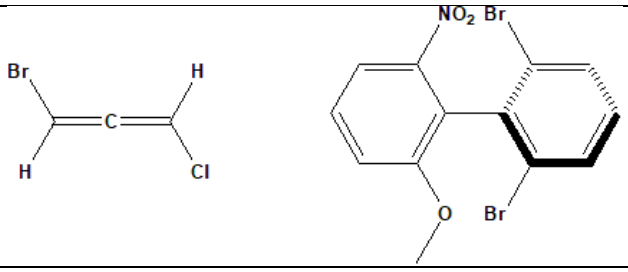
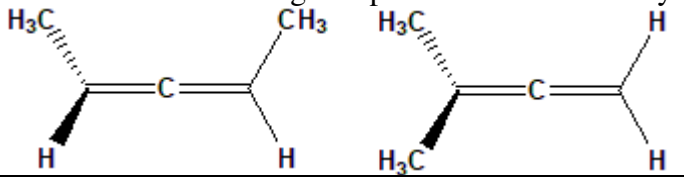
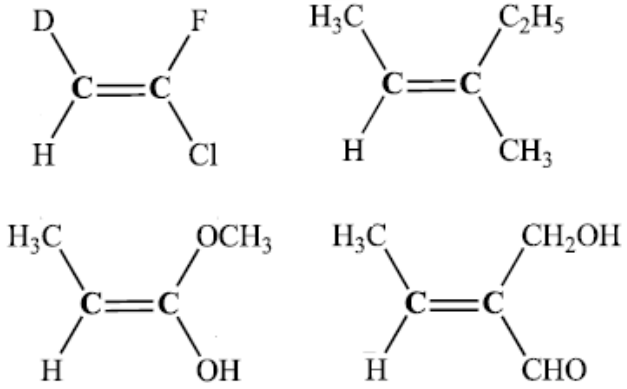
1	43.	Designate E/Z notation and name the following compounds i)  ii) 	2
1	44.	What are erythrose and threoseconfigurations?	2
1	45.	Write the cis- and trans- isomers of 1,2-dimethylcyclopropane	2
2	46.	Give the Arrhenius concept of acids and bases withexamples.	**2**
2	47.	Give the Bronsted-Lowry concept of acids and bases withexamples.	**2**
2	48.	Between methylamine and ammonia, which is more basic and why?	**2**
2	49.	Aniline is a weaker base than ammonia. Justify the statement.	**2**
2	50.	Give the Lewis concept of acids and bases withexamples.	**2**
2	51.	Between aniline and nitro aniline, which is more basic and why?	**2**
2	52.	What is meant by hyperconjugation effect? Give suitable example.	**2**
2	53.	Define Inductive effect? Give an example.	**2**
2	54.	What is Inductive effect? Give an example.	**2**
2	55.	Between phenol and nitro phenols which is more acidic and why?	**2**
2	56.	Arrange the following in decreasing order of their acid strength: P-cresol, phenol and p-chlorophenol. Give reason for your answer.	**2**
2	57.	Between o-nitro phenol and p-nitro phenol which is more acidic and why?	**2**
2	58.	Between aniline and P-toluidine, which is more basic and why?	**2**
2	59.	Which is more basic and why? (i) $H_2N\text{-}CH_2\text{-}COO^-$ (ii) $H_2N\text{-}CH_2\text{-}COOCH_3$	**2**
2	60.	P-nitro benzoic is more acidic than P-hydroxy benzoic and justify the statement.	**2**
2	61.	Rank the following compounds in the order of acidity and give reason. $Cl\text{-}CH_2\text{-}COOH$, $CH_3\text{-}COOH$ and HCOOH.	**2**
2	62.	Rank the following compounds in the order of acidity. $Br\text{-}CH_2\text{-}COOH$, $Cl\text{-}CH_2\text{-}CH_2\text{-}COOH$, $F\text{-}CH_2\text{-}COOH$, $I\text{-}CH_2\text{-}CH_2\text{-}COOH$, $CH_3\text{-}COOH$.	**2**
2	63.	Chloroacetic acid is stronger acid than acetic acid. Give reason.	**2**
2	64.	Rank the following compounds in the order of acidity, HCOOH, $CH_3\text{-}CH_2\text{-}CH_2\text{-}COOH$, CH_3COOH, $CH_3\text{-}CH_2\text{-}COOH$	**2**
2	65.	Phenols are acidic than alcohols. Why?	**2**
2	66.	What is hydrogen bonding? Give its types.	**2**

2	67.	What is an intermolecular and intramolecular hydrogen bonding?	**2**
2	68.	Carboxylic acids are stronger acid than alcohol. Justify the statement.	**2**
2	69.	Rank the following compounds in the order of acidity, CH_3OH, CH_3NH_2, CH_3CH_3	**2**
2	70.	Mention the types of reaction mechanism with one example each.	**2**
2	71.	Define reaction mechanism and mention its importance.	**2**
2	72.	What is reaction mechanism?and Mention its classification.	**2**
2	73.	Give the classification of reaction mechanism	**2**
2	74.	Give the importance of reaction mechanism.	**2**
2	75.	Give two examples for neighbouring group participation.	**2**
2	76.	What is SN^1 reaction? Give an example.	**2**
2	77.	What is SN^ireaction? Give an example.	**2**
2	78.	Give the difference between SN^1 and SN^2 reaction.	**2**
2	79.	Differentiate SN^1 and SN^2 reaction.	**2**
2	80.	What is SN^2 reaction? Give an example.	**2**
2	81.	How is benzyne generated?	**2**
2	82.	What is non-classical carbocation? Give example.	**2**
2	83.	What are carbanions? Give example.	**2**
2	84.	What are free radicals? Give example.	**2**
2	85.	What are carbenes? How carbene generated?	**2**
2	86.	What are nitrenes? Give any one method for their generation.	**2**
2	87.	How nitrile oxide generated?	**2**
2	88.	What are nitrile ylides? Give example.	**2**
3	89.	Write the preparation of pyrrole.	**2**
3	90.	Write any two physical properties of pyrrole.	**2**
3	91.	Pyrrole is not a weak base also acts as very weak acid. Give reasons.	**2**
3	92.	Why pyrrole acts as weak base furan show basic character. Justify	**2**
3	93.	Explain the preparation of pyrrole by paulknorr method.	**2**
3	94.	Why electrophilic substitution reaction occurs at C-2 Position in furan?	**2**
3	95.	Name the reagents used for nitration and sulphonation of pyrrole	**2**
3	96.	Electrophilic substitution reaction of pyrrole does not carry in presence of strong acid. Give reason.	**2**

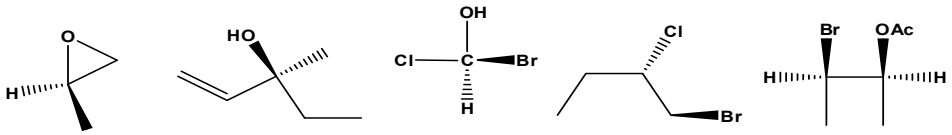
3	97.	Write the resonance structure of pyrrole.	**2**
3	98.	How to confirm the hybrid character of furan?	**2**
3	99.	Write the preparation of furan.	**2**
3	100.	Write any two physical properties of furan	**2**
3	101.	Furan show basic character. Justify	**2**
3	102.	Explain the preparation of furan by dry distillation method.	**2**
3	103.	Why electrophilic substitution reaction occurs at C-2 Position in furan?	**2**
3	104.	Name the reagents used for nitration and sulphonation of furan.	**2**
3	105.	Nitration of furan does not carry in presence of strong acid. Give reason.	**2**
3	106.	Milder condition is required for the synthesis mono halogenated derivative of furan. Justify	**2**
3	107.	Write the resonance structure of furan.	**2**
3	108.	How to confirm the hybrid character of furan?	**2**
3	109.	Write the structure of thiphene and purine	**2**
3	110.	How thiophene can be separated from benzene?	**2**
3	111.	How thiohene can be synthesis from acetylene?	**2**
3	112.	Write any two preparation of thiophene.	**2**
3	113.	Write the commercial method for the preparation of thiophene.	**2**
3	114.	Write the resonance structure of thiophene,	**2**
3	115.	Give any two physical properties of thiophene.	**2**
3	116.	Thiophene is an aromatic in nature. Justify	**2**
3	117.	Write structure of indole and pyridine	**2**
3	118.	Explain the preparation of indole by Lipp synthesis.	**2**
3	119.	What is fact behind the name indole?	**2**
3	120.	Write the resonance structure of indole.	**2**
3	121.	Mention any two physical properties of indoles.	**2**
3	122.	How electrophillic attack at C-3 is favourable then C-2 attack in indole.	**2**
3	123.	Pyridine is stronger base than pyrrole. Justify	**2**
3	124.	Write the preparation of pyridine from piperidine.	**2**
3	125.	How pyridine can be synthesis from acetylene?	**2**
3	126.	Write the commercial method for the synthesis of pyridine.	**2**

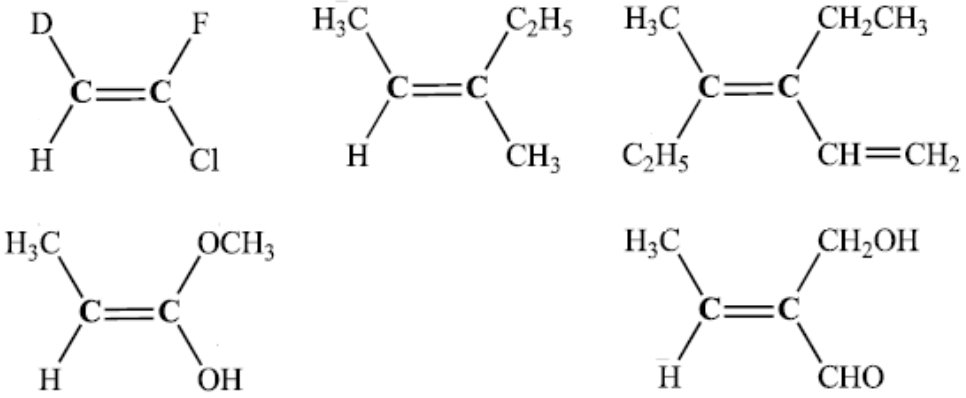
3	127.	Write the bromination reaction of pyridine.	**2**
3	128.	Write the oxidation and reduction reaction of pyridine.	**2**
3	129.	Write the structure of quinoline and pyridine.	**2**
3	130.	Write the structure of isoquinoline and pyrrole.	**2**
3	131.	Write any two physical properties of quinoline.	**2**
3	132.	Write the resonance structure of quinoline.	**2**
3	133.	How isoquinoline can be synthesis from cinnamaldehyde.	**2**
3	134.	Write the oxidized product, when isoquinoline is treated with $KMnO_4$.	**2**
3	135.	Write the structure of pyrazole and pyrimidine.	**2**
3	136.	Write the structure of pyrimidine and imidazole.	**2**
3	137.	Name the amino acid and vitamin where imidazole ring is present.	**2**
3	138.	Write the synthesis of imidazoles from alkohalketones.	**2**
3	139.	How imidazole can be synthesis from 1,3 dipolar cyclo addition reaction?	**2**
3	140.	Imidazole have higher melting point than pyrazole. Justify	**2**
3	141.	Imidazole is stronger base than pyrrole. Give reasons.	**2**
3	142.	Imidazole is an aromatic heterocyles. Give reasons	**2**
3	143.	What are uses of pyrazoles?	**2**
3	144.	Name some market drug which contain pyrazole ring.	**2**
3	145.	Write the structure of pyrone and furon.	**2**
3	146.	What are pyrones?	**2**
3	147.	Write the structure of 2H-Pyrone and 4H-Pyrone	**2**
3	148.	Write the resonance structure of Pyrones.	**2**
3	149.	What happen when pyrones undergo reduction with Nickel?	**2**
3	150.	Write the structure of Coumarin and furan	**2**
3	151.	Write the structure of Chromene and pyrimidine	**2**
3	152.	What are coumarin?	**2**
3	153.	What are chromenes?	**2**
3	154.	Write the bromination of coumarins?	**2**
3	155.	How chromene react with Grignard reagents?	**2**
3	156.	Write the structure of Pyridine and Pyrimidine	**2**
3	157.	Give the difference between Pyridine and Pyrimidine	**2**

3	158.	"Pyrimidine is an aromatic heterocylcle" Give reasons	**2**
3	159.	"Pyrimidine is less reactive towards electrophilic substitution reaction". Justify	**2**
3	160.	Pyrimidine is used in pharmacological industry. Give reasons	**2**
3	161.	Give a method for alkylation of pyrimidine	**2**
3	162.	Write the structure of 5-fluorouracil and mention one use.	**2**
3	163.	Write the structure of Purine and thiophene	**2**
3	164.	How purine is different from pyrimidine	**2**
3	165.	Write the structure of purine with its hetero atom position number.	**2**
3	166.	"Purine is aromatic in nature" Give reasons	**2**
3	167.	Write the structure of Purine, hence write the number delocalized π-electron.	**2**
3	168.	Write the tautomerism of Purine.	**2**
3	169.	Discuss the Trauba synthesis of purine.	**2**
3	170.	Write structure of 9H-Purine and 7H-Purine	**2**
1	171.	A certain amount (y gm) of an optically pure compound shows a rotation of $+60^{o}$ when dissolved in a methanol (10 mL) kept in a cell of path length 100 mm. The specific rotation of the compound in the same solvent under similar conditions (wavelength and temperature) is $+ 120^{0}$. Find the value of y.	4
1	172.	Draw the four stereoisomers of 4-methyl-2-hexanol and give the relationship of each other's.	4
1	173.	Discuss optical isomerism in biphenyls.	4
1	174.	Discuss optical isomerism in Allenes.	4
1	175.	Write a note resolution of racemates by any two methods.	4
1	176.	Explain stability of cis and trans decalin	4
1	177.	Write a note on Optical purity and enantiomeric excess.	4
1	178.	Write all stereoisomers of 2-bromo-3acetoxy butane and assign the configuration.	4
1	179.	Write the stable conformation of 1,2-dimethyl cyclohexane and justify.	4
1	180.	Write the stable conformation of 1,3-dimethyl cyclohexane and justify.	4
1	181.	Write the stable conformation of 1,4-dimethyl cyclohexane and justify.	4
1	182.	Assign R/S configuration for	4

			
1	183.	Explain stereo selective and stereo specific reactions with example.	4
1	184.	Write a note on E/Z notations.	4
1	185.	Explain R and S notation taking suitable example.	4
1	186.	Mention the methods available for determination of Geometrical isomers. Explain one method.	4
1	187.	Describe the conformation of Decalins.	4
1	188.	Which of the following compound is Chiral? Justify. 	4
2	189.	Designate E/Z notation and name the following compounds 	4
2	190.	Explain the structural effect on the strength of acids.	4
2	191.	Describe the structural effect on the strength of bases.	4
2	192.	Describe the structural effect on the strength of acids.	4
2	193.	Explain the structural effect on the strength of bases.	4
2	194.	What is negative and positive Inductive effect? Give its applications.	4
2	195.	What is negative and positive Mesomeric effect? Give its applications.	4
2	196.	Write a note on Mesomeric effect.	4
2	197.	What is negative and positive Mesomeric effect Give its applications.	4
2	198.	Write a brief note on hyperconjugation.	4
2	199.	What is hydrogen bonding? Elaborate its types.	4

2	200.	What is hydrogen bonding? Explain its types.	4
2	201.	Explain hyperconjugation with suitable examples.	4
2	202.	What is hyperconjugation? Explain it with examples.	4
2	203.	Write a note on hyperconjugation.	4
2	204.	Write a note on Inductine effect.	4
2	205.	What is hydrogen bonding? Explain the effect of hydrogen bonding on acidity of nitrophenols.	4
2	206.	Write a note on importance of reaction mechanism.	4
2	207.	Explain the classification of reaction reaction with examples.	4
2	208.	Describe the classification of organic reactions with example.	4
2	209.	Explain the kinetics and mechanism of SN^2 reaction.	4
2	210.	Describe the kinetics and mechanism of SN^1 reaction.	4
2	211.	Describe the kinetics and mechanism of SN^2 reaction.	4
2	212.	Discuss with example SN^i reaction.	4
2	213.	Explain with example SN^1 reaction.	4
2	214.	Describe with example SN^2 reaction.	4
2	215.	Write a brief note on SN^i reaction.	4
2	216.	Illustrate the effect of neighboring group participation during the nucleophilic substitutions with suitable example.	4
2	217.	What is neighboring group participation? Explain it with suitable example.	4
2	218.	Discuss briefly the benzyne mechanism.	4
2	219.	With suitable mechanism explain Bucherer reaction.	4
2	220.	Write a brief account of benzyne mechanism.	4
2	221.	Discuss the mechanism and stereochemical factors affecting the rate of SNi reaction.	4
2	222.	Explain the mechanism and stereochemistry of SN1 reaction with suitable example.	4
2	223.	Discuss the mechanism and stereochemical factors affecting the rate of SN^2 reaction.	4
2	224.	Illustrate the mechanism involved in Friedle Craft's reaction.	4
2	225.	Discuss the mechanism involved in halogenations reaction.	4
2	226.	Explain Mannich reaction with suitable example.	4
2	227.	Describe Mannich reaction.	4
2	228.	With example explain Mannich reaction.	4

2	229.	Discuss the formation, stability and reactions of carbocation.	4
2	230.	Discuss the structure and stability of carbanions. Give any one reaction.	4
2	231.	Discuss the reactivity and stability carbocation.	4
2	232.	Discuss the generation and stability of carbanion.	4
2	233.	Explain the generation and stability of free radicals.	4
2	234.	Elaborates the generation and stability of carbocations.	4
2	235.	What are ylides? Give its synthetic applications.	4
2	236.	Discuss the methods of trapping carbene and nitrene with examples.	4
2	237.	What are nitrile oxides? How are they generated? Give ant two synthetic applications of them	4
2	238.	What are nitrenes? How are they generated? Give examples to state their stability.	4
2	239.	Write a note on nitrile ylides.	4
1	240.	Assign a R or S configuration for each stereo-center: 	5
1	241.	A sample of pure (S)-2-butanol was placed in a 10.0 cm polarimeter tube. Using the D line of a sodium lamp, the observed rotation at 20°C was $\alpha = +104^{o}$. The density of this compound is 0.805 g ml^{-1}. What is the specific rotation of (S)-2-butanol? Give the structures of (S)-2-butanol and (R)-2-butanol	5
1	242.	Predict the number of stereoisomers possible for aldopentose. Write all stereoisomers and envisage the absolute configuration of each chiral carbon.	5
1	243.	Discuss the effect of conformation on the course of rate of reaction with suitable examples.	5
1	244.	Discuss the conformation analysis of n-butane with energy profile diagram.	5
1	245.	Discuss the conformation analysis of ethane with energy profile diagram.	5
1	246.	Designate E/Z notation and name the following compounds	5

			
1	247.	Write a note on: i. Resolution through formation of Diastereomers. ii. Resolution employing Microorganism.	5
2	248.	Write a note on inductive effect taking suitable examples.	5
2	249.	Elaborate hyperconjugation with suitable examples.	5
2	250.	Write a note on applications of hyperconjugation.	5
2	251.	Explain the structural effect on the strength of acids.	5
2	252.	What is Inductive effect? Expline its types with suitable example.	5
2	253.	Write a note on Inductine effect.	5
2	254.	What is hyperconjugation? Write a note on applications of hyperconjugation.	5
2	255.	What is Inductive effect? Discuss its types with suitable example.	5
2	256.	What is Mesomeric effect? Expline its types with suitable example.	5
2	257.	Write a note on Mesomeric effect.	5
2	258.	What is Mesomeric effect? Discuss its types with suitable example.	5
2	259.	Describe the structural effect on the strength of bases.	5
2	260.	What is hydrogen bonding? Elaborate its types.	5
2	261.	What is hydrogen bonding? Explain its types.	5
2	262.	Describe the structural effect on the strength of acids.	5
2	263.	Explain the structural effect on the strength of bases.	5
2	264.	What is reaction mechanism? Write a note on importance of reaction mechanism.	5
2	265.	Define reaction mechanism? Explain the classification of reaction reaction with examples.	5
2	266.	What is reaction mechanism? Describe the classification of organic reactions with example.	5

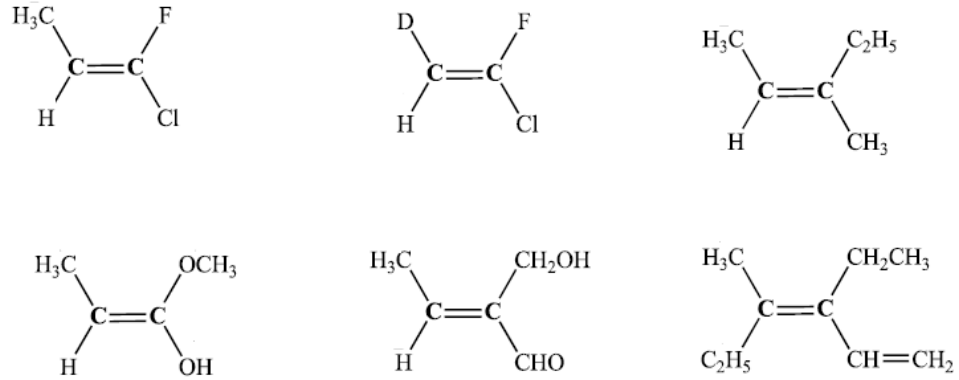
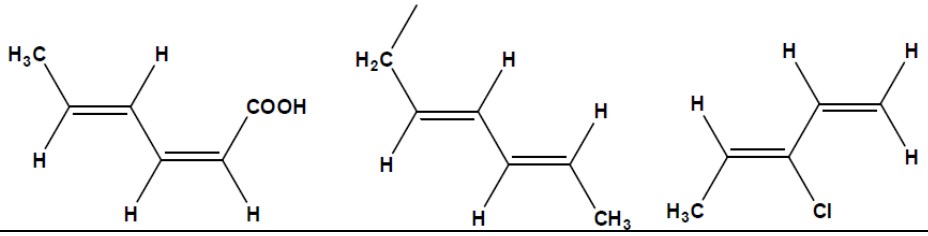
2	267.	Explain with example SN^1 reaction.	5
2	268.	Describe with example SN^2 reaction.	5
2	269.	Write a brief note on SN^ireaction.	5
2	270.	Discuss SN^i mechanism with suitable example.	5
2	271.	Describe the kinetics and mechanism of SN^2 reaction with a suitable example.	5
2	272.	Explain the kinetics and mechanism of SN^1 reaction with a suitable example.	5
2	273.	Illustrate the effect of neighboring group participation during the nucleophilic substitutions with suitable example.	5
2	274.	With suitable mechanism explain Bucherer reaction.	5
2	275.	Explain Bucherer reaction.	5
2	276.	Write a brief account of benzyne mechanism.	5
2	277.	Discuss the mechanism and stereochemical factors affecting the rate of SNi reaction with suitable example.	5
2	278.	Explain the mechanism and stereochemistry of SN^1 reaction with suitable example.	5
2	279.	Discuss the mechanism and stereo chemical factors affecting the rate of SN^2 reaction with suitable example.	5
2	280.	Explain the following reaction with mechanism: (i) Friedel-Crafts alkylation (ii) Halogenation	5
2	281.	Explain the following reaction with mechanism: (i) Friedel-Crafts acylation (ii) Sulphonation	5
2	282.	Explain the following reaction with mechanism: (i) Sulphonation (ii) Halogenation	5
2	283.	Discuss the reaction and mechanism of (i) Mannich reaction (ii)Vilsmeier-Haack reaction.	5
2	284.	Explain the following reaction with mechanism: (i) Chloromethylation (ii) Vilsmeier-Haack reaction	5
2	285.	Discuss the reaction and mechanism of (i)Chloromethylation(ii)Vilsmeier-Haack reaction.	5
2	286.	Elaborate the formation, stability and reactions of carbocations	5
2	287.	Discuss the structure and stability of carbanions.Give any one reaction.	5
2	288.	What are carbocations and carboanions? Discuss their stability.	5
2	289.	Discuss the generation and stability of carbanion.	5
2	290.	Explain the generation and stability of free radicals.	5
2	291.	Elaborates the generation and stability of carbocations.	5
2	292.	What are nitrile oxides? How are they generated? Give their any two synthetic applications.	5
2	293.	What are nitrenes? How are they generatd? Discuss their stability.	5
2	294.	What are carbenes? How are they generated? Give their any two synthetic	5

		applications.	
2	295.	Write a note on nitrile ylides.	5
2	296.	How nitrile imines are synthesized? Give their applications.	5
3	297.	Discuss the formation of sigma and pi bond in furan and pyrrole ring.	5
3	298.	Write any five electrophilic reaction of furan.	5
3	299.	Summarize a report on preparation, structure and reaction of furan.	5
3	300.	Write any two preparation of furan.	5
3	301.	Explain the following reaction of furan i) Ring opening reaction ii) Diels –Alder reaction iii) Reduction	5
3	302.	Write the reaction of i) Furan with mercuric chloride ii) Furan with n-butyl Lithium iii) Freidel Crafts Acylation	5
3	303.	Explain the Paul Knoor synthesis of pyrrole with mechanism.	5
3	304.	Write the preparation and chemical reaction of pyrrole.	5
3	305.	Summarize the report on the structure and chemical reaction of pyrrole.	5
3	306.	Explain structure, acidic and basic character of pyrrole.	5
3	307.	Discuss electrophillic substitution reaction of pyrrole.	5
3	308.	Write a note on preparation, structure and chemical reaction of pyrrole.	5
3	309.	Write the following reaction of indole i) Reimer-Tiemannformylation ii) Friedel Craft's acylation iii) Kolbe-Schmitt carboxylation iv) Ring expansion reaction v) Ring opening reaction	5
3	310.	Write the product when thiophene undergoes reduction.	5
3	311.	Write a note on electrophhlic reaction substitution reaction of thiophene.	5
3	312.	Summarize a report on the preparation and electrophillic reaction of thiophene.	5
3	313.	Discuss the preparation, structure and chemical reaction of thiophene.	5
3	314.	Write any five preparation of thiophene. Hence, write its resonance structure.	5

3	315.	Explain the Fisher indole synthesis with its mechanism	5
3	316.	Write the synthesis of indole by Reissert method.	5
3	317.	Discuss the structure and physical properties of Indole.	5
3	318.	Explain the electrophillic substitution reaction of inodole.	5
3	319.	Write the following reaction of indole i) Reimer-Tiemannformylation ii) Friedel Craft's acylation iii) Diazo coupling iv) Mannich Reaction v) Reduction	5
3	320.	Summarize a report on the synthesis and reaction of indole.	5
3	321.	How electrophillic attack at C-3 is favourable then C-2 and C-4 attack in indole.	5
3	322.	Write the resonance structure of pyridine.	5
3	323.	Explain the electrophillic substitution reaction of pyridine in detail.	5
3	324.	Write a note on electrophhilic attack at C-2, C-3 and C-4 position of pyridine and hence write nitration and sulfonation reaction.	5
3	325.	Discuss the nucleophillic substitution reaction of pyridine with an example.	5
3	326.	Write the following reaction of pyridine with mechanism i) Reaction with sodamide ii) Reaction with n-butyllithium.	5
3	327.	Discuss the preparation and structure of pyridine.	5
3	328.	Summarize a report on the synthesis, structure and chemical reaction of pyridine.	5
3	329.	Discuss the skraup synthesis of quinoline with its mechanism.	5
3	330.	Explain the structure oquinoline and hence also write its resonance structure.	5
3	331.	Write a note on electrophillic andnucleophillic substitution reaction of quinoline.	5
3	332.	Explain the Bischler-Napierrlski synthesis of isoquinoline.	5
3	333.	Summarize a report on the synthesis, reactivity and chemical reaction of quinoline.	5
3	334.	Write the reaction when i) Quinoline react with Acid ii) Quinoline react n-butyl lithium	5

		iii) Isoquinoline undergo oxidation iv) Isoquinoline undergo reduction v) Quinoline react with alkyl hakides	
3	335.	Write a note on synthesis and oxidation reaction of isoquinoline.	5
3	336.	Explain the synthesis of imidazole stating from i) Diketone with ammonia and ii) formamide	5
3	337.	Explain the physicals and chemical properties of imidazoles	5
3	338.	Discuss electophillic substitution reaction of imidazoles.	5
3	339.	What happen when imidazoles undergo i) Mercuration ii) Protonation iii) Bromination iv) oxidation v) Reduction	5
3	340.	Summarize a report on synthesis and chemical properties of imidazoles.	5
3	341.	Explain the synthesis of pyrozole starting from β-Ketones	5
3	342.	What are pyrazoles? Discuss the synthesis of pyrazoles from diazomethane	5
3	343.	Summarize a report on the synthesis and reaction of pyrazole.	5
3	344.	Write a note on synthesis, reactivity and chemical reaction of pyrazole.	5
3	345.	What are pyrones? How it is synthesizied?	5
3	346.	What are pyrones? How pyrones can be synthesis from maleic acid?	5
3	347.	Discuss the electrophillic and nucleophilic substitution reaction of pyrones	5
3	348.	Summarize a report on synthesis and reactivity of Pyrones	5
3	349.	Write a note on synthesis and reaction of pyrones	5
3	350.	What are chromenes? How it is synthesized?	5
3	351.	What are coumarins? How it is synthesized?	5
3	352.	Explain the synthesis of coumarins from o-hydroxybenzaldhydes?	5
3	353.	Summarize a report on synthesis and reaction of Coumarin	5
3	354.	Write a note on synthesis and reaction of Chromenes	5
3	355.	Explain the following reaction of Coumarin i) Reduction	5

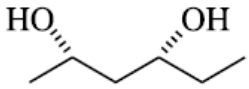
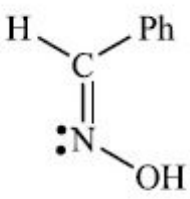
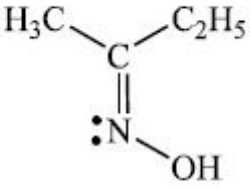
		ii) Electrophillic substitution reaction iii) Nucleophillic substitution reaction iv) with Grignard Reagents	
3	356.	Explain the following reaction of Chromenes i) Reduction ii) Electrophillic substitution reaction iii) Nucleophillic substitution reaction iv) with Grignard Reagents	5
3	357.	Discuss the synthesis of chromenes from i) Aromatic aldehyde ii) Phenol and β-ketoenols	5
3	358.	Descirbe the synthesis of pyrimidine from i) Unsaturated Ketone ii) 2,4-dichloropyrimidine	5
3	359.	Discuss the nucleophilic substitution reaction of Pyrimidine	5
3	360.	Discuss the electrophilic substitution reaction of Pyrimidine	5
3	361.	Explain the synthesis of pyrimidine form malonic ester.	5
3	362.	Write a note on synthesis and reactivity of Pyrimidine	5
3	363.	Summarize a report on reaction of Pyrimidine.	5
3	364.	Explain the synthesis purine by Fisher method	5
3	365.	Write the preparation of Purine from i) Imidazole ii) Formaide	5
3	366.	Discuss the reaction of synthesis and reaction of purine	5
3	367.	Discuss the nucleophilic substitution reaction of purine	5
3	368.	Discuss the electrophilic substitution reaction of purine	5
3	369.	Write a note on synthesis and reactivity of Purine	5
3	370.	Summarize a report on reaction of Purine	5
3	371.	Discuss the following reaction i) Reaction of purine with acid and hence mention the predominant product. ii) Reaction of purine with CH_3I and with H_2O_2	5
1	372.	Write a note resolution of racemates by any three methods.	6

1	373.	Write a note on flipping of cyclohexane	6
1	374.	Describe the determination of configuration of geometrical isomers by physical and chemical methods.	6
1	375.	Discuss the conformation analysis of 1, 2 and 1, 4 – disubstituted cyclohexane.	6
1	376.	Discuss the conformation analysis of 1, 3 and 1, 4 – disubstituted Cyclohexane.	6
1	377.	Discuss the conformation analysis of 1, 2 and 1, 3 – disubstituted Cyclohexane.	6
1	378.	Discuss the conformation analysis of mono substitutedand 1, 4 – disubstituted Cyclohexane.	6
1	379.	Write all the possible stereoisomers for Aldotetroses and identify R/S; D/L and Threose/Erythrose configuration.	6
1	380.	Designate E/Z notation and name the following compounds 	6
1	381.	Write a note on: i. Resolution through formation of Diastereomers. ii. Resolution employing Enzymes.	6
1	382.	1. Write a note on structure and hybridization in Allenes. 2. Write enantiomers of 6,6'-dinitrobiphenyl-2,2'-dicarboxylic acid.	6
1	383.	Designate E/Z notation and name the following compounds 	6
2	384.	Give a brief note on inductive effect taking suitable examples.	6
2	385.	Explain the structural effect on the strength of acids.	6
2	386.	Describe the structural effect on the strength of bases.	6
2	387.	Describe the structural effect on the strength of acids.	6
2	388.	Explain the structural effect on the strength of bases.	6
2	389.	What is Inductive effect? Explain its types and applications.	6

2	390.	Write a note on Inductive effect.	6
2	391.	What is hyperconjugation? Write a note on applications of hyperconjugation.	6
2	392.	What is Inductive effect? Discuss its types and applications.	6
2	393.	What is Mesomeric effect? Explain its types with suitable example.	6
2	394.	Write a note on Mesomeric effect.	6
2	395.	What is Mesomeric effect? Discuss its types with suitable example.	6
2	396.	What is hydrogen bonding? Explain its types.	6
2	397.	What is hydrogen bonding? Elaborate its types.	6
2	398.	What is hyperconjugation? Write a note on applications of hyperconjugation.	6
2	399.	What is reaction mechanism? Write a note on importance and classification of reaction mechanism.	6
2	400.	Define is reaction mechanism? Write a note on importance and classification of reaction reaction with examples.	6
2	401.	What is reaction mechanism? Describe the importance and classification of organic reactions with example.	6
2	402.	Define reaction mechanism? Describe the importance and classification of organic reactions with example.	6
2	403.	Discuss the mechanism and stereo chemical factors affecting the rate of SN2 reaction.	6
2	404.	Describe the kinetics and mechanism of SN^2 reaction.	6
2	405.	Explain the kinetics and mechanism of SN^1 reaction with a suitable example.	6
2	406.	Discuss the mechanism and stereo chemical factors affecting the rate of SN^ireaction with suitable example.	6
2	407.	Discuss the mechanism and stereo chemical factors affecting the rate of SNi reaction with suitable example.	6
2	408.	Explain the mechanism and stereochemistry of SN^1 reaction with suitable example.	6
2	409.	Discuss the mechanism and stereo chemical factors affecting the rate of SN^2 reaction with suitable example.	6
2	410.	Explain the following reaction with mechanism: (i) Friedel-Crafts alkylation (ii) Halogenation	6
2	411.	Explain the following reaction with mechanism: (i) Friedel-Crafts acylation (ii) Sulphonation	6
2	412.	Explain the following reaction with mechanism: (i) Sulphonation (ii) Halogenation	6
2	413.	Discuss the reaction and mechanism of (i) Mannich reaction (ii)Vilsmeier-Haack reaction.	6
2	414.	Explain the following reaction with mechanism: (i) Chloromethylation (ii) Vilsmeier-Haack reaction	6
2	415.	Discuss the reaction and mechanism of (i)Chloromethylation(ii)Vilsmeier-Haack reaction.	6

2	416.	What are carbocations and carboanions? Discuss their stability.	6
2	417.	Elaborate the formation, stability and reactions of carbocations	6
2	418.	Discuss the structure and stability of carbanions. Give any one reaction.	6
2	419.	What are carbocations and carboanions? Discuss their stability.	6
2	420.	Discuss the generation and stability of carbanion.	6
2	421.	Explain the generation and stability of free radicals.	6
2	422.	What are nitrile oxides? How are they generated? Give their any two synthetic applications.	6
2	423.	What are nitrenes? How are they generatd? Discuss their stability.	6
2	424.	What are carbenes? How are they generatd? Discuss their stability.	6
2	425.	How arynes are synthesized? Discuss their stability and give their applications.	6
2	426.	How nitrile imines are synthesized? Give their applications.	6
2	427.	Write a note on nitrile ylides.	6

Question Paper Pattern- Model Question Paper

St. Philomena's College (Autonomous), Mysore			
I Semester –Course M.Sc Final Examination			
Subject : Chemistry			
Title: STEREOCHEMISTRY, REACTION MECHANISMS AND HETEROCYCLIC CHEMISTRY			
Time: 3 hrs			**Max Marks: 70**
PART-A			
1		**Answer any ten of the following:**	**10x2=20**
	a	Identify the chiral centers and provide the absolute configuration (R/S) of the given molecule. 	2
	b	What are stereoselective reaction? Give an example.	2
	c	Draw the Sawhorse and Newman Projection formulae of 2-bromo-3-hydroxy-butane.	2
	d	Designate E/Z notation and name the following compounds i)  ii) 	2
	e	Give the Bronsted-Lowry concept of acids and bases wthexapmles.	2
	f	Give the classification of reaction mechanism	2
	g	Rank the following compounds in the order of acidity and give reason. Cl-CH_2-COOH, CH_3-COOH and HCOOH.	2
	h	What is nitration? Give example	2
	i	Why pyrrole acts as weak base furan show basic character. Justify	2
	j	Write the commercial method for the synthesis of pyridine.	2
	k	Write the structure of thiphene and purine	2
	l	Write the preparation of furan.	2
PART-B			
		Answer any Five of the following:	**5 x 10= 50**

2	a	Describe the determination of configuration of geometrical isomers by physical and chemical methods.	6
	b	Explain stability of cis and trans decalin.	4
3	a	Discuss the conformation analysis of n-butane with energy profile diagram.	5
	b	Predict the number of stereoisomers possible for aldopentose. Write all stereoisomers and envisage the absolute configuration of each chiral carbon.	5
4	a	Write a note on: (i) Resolution through formation of Diastereomers. (ii) Resolution employing Microorganism.	5
	b	What are carbocations and carboanions? Discuss their stability.	
5	a	Explain the structural effect on the strength of acids.	6
	b	Write a note on nitrile ylides.	4
6	a	Discuss the mechanism and stereochemical factors affecting the rate of SN2 reaction with suitable example.	5
	b	Explain the following reaction with mechanism: (i) Friedel-Crafts acylation (ii) Sulphonation	5
7	a	Write the reaction of i) Furan with mercuric chloride ii) Furan with n-butyl Lithium iii) Freidel Crafts Acylation	5
	b	Write the product when thiophene undergoes reduction.	5
8	a	What are pyrones? How pyrones can be synthesis from maleic acid?	5
	b	Summarize a report on the synthesis, reactivity and chemical reaction of quinoline.	5

Stereochemistry, Reaction Mechanisms and Heterocyclic Chemistry